

Final Exam Key (Confidential)

Q1: Encapsulation & Overhead

(a) Frame Sizes (Data + 62 bytes overhead):

- Message A: **262 bytes**
- Message B: **862 bytes**
- Message C: **1462 bytes**

(b) Total Overhead:

- **186 bytes** (62×3)

(c) Average Overhead Percentage:

- *Note: Full marks were awarded for any of the following valid calculation methods:*
 1. **7.75%** ($\text{Overhead} \div \text{Data Payload} \rightarrow 186/2400$)
 2. **7.19%** ($\text{Overhead} \div \text{Total Frame Size} \rightarrow 186/2586$)
 3. **11.7%** (Average of individual packet percentages)

Q2: Application Layer Protocols

- i) **Set-Cookie: 6001** (or **ID=6001**)
- ii) **Cookie: 6001**
- iii) **HTTP** (or **HTTPS**)
- iv) **SMTP**
- v) **Push**

Q3: Internet Checksum

- **(a) Computed Checksum: 2A24**
 - Calculation: $5366 + 74B4 = C81A \rightarrow C81A + 0DC1 = D5DB$.
 - 1s Complement of $D5DB = 2A24$.
- **(b) Error Detection:**
 - Receiver Sum: $C79A + 0DC1 + 2A24$ (Checksum) = $FF7F$.
 - Since the result is **not** $FFFF$ (all 1s), the error is **detected**.

Q4: TCP Congestion Control (Reno)

(a) Completion of Table 4 Note: RTT 5 is transmitted under CA rules. The sender detects the Timeout at the start of RTT 6.

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RTT	cwnd	ssthresh	ACK Rcvd	Segments Transmitted	Timer	Status	Mode
0	0	12	N/A	N/A	600	No Loss	N/A
1	1	12	N/A	0	600	No Loss	SS
2	2	12	100	100, 200	600	No Loss	SS
4	8	12	700	700 ... 1200	600	No Loss	SS
5	12	12	1300	1300 ... 1800	600	No Loss	CA
6	1	6	1300	1300	1200	Timeout	SS
10	7	6	2600	2600 ... 3200	1200	No Loss	CA

(b) Sequence number of lost segment: 1300

Q5: Network Layer

(a) Longest Prefix Match:

- Packet destined for `200.23.16.20` matches the entry for `...16.16/28`.
- **Answer: Interface 6**

(b) Distance Vector (Count-to-Infinity):

- i) New Cost to X: **9** (Route via Y: $1 + 8$)
- ii) Traceroute path: **X, Y, X, Y, X...** (Looping)
- iii) Value of TTL when packet dropped: **0 or 1** (Both accepted)

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Q6: BGP Policy

(a) Can Z implement this policy?

- **Yes**

(b) How? (Any two of the following):

1. **Import Filtering:** Z configures its router to reject advertisements from Y that contain "AS X" in the AS-Path.
2. **Community Attributes:** Z tags its advertised routes to Y with a `NO_EXPORT` (or similar) community, telling Y not to propagate these routes to X.
3. **Broken Return Path:** If Z refuses to accept routes to X, it cannot send return traffic, effectively isolating X.

Q7: Slotted ALOHA

- Parameters: $N = 4$ nodes, Transmission Probability $p = 0.2$.

(a) Probability that Node A succeeds in Slot 1:

- Logic: A transmits (p) AND B, C, D do not ($1 - p$ each).
- Calculation: $0.2 \times (0.8)^3 = \mathbf{0.1024}$ (or 10.24%)

(b) Probability that Node A succeeds for the first time in Slot 4:

- Logic: [A fails Slot 1] $\times$ [A fails Slot 2] $\times$ [A fails Slot 3] $\times$ [A succeeds Slot 4].
- Probability of A failing: $1 - P(\text{success}) = 1 - 0.1024 = 0.8976$.
- Calculation: $(0.8976)^3 \times 0.1024 \approx \mathbf{0.074}$ (or 7.4%)

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Q8: MAC Addresses & Switching

- i(a): Destination MAC: **11:11:11:11:11:11** | Protocol: **ARP**
- i(b): Source: **AA:AA:AA:AA:AA:AA** $\rightarrow$ Dest: **11:11:11:11:11:11**
- i(c): Source: **33:33:33:33:33:33** $\rightarrow$ Dest: **BB:BB:BB:BB:BB:BB** (Source is R1's interface)
- ii: Destination MAC: **DD:DD:DD:DD:DD:DD**
- iii: **No** (Routers do not forward broadcast traffic).